

Paper Information Record — Greene2021

Paper ID: 0019 | Oxford Missing Data Study

General Information

Field	Value
Paper ID	0019
Title	Anti-Discrimination Laws and Firm Performance
Authors	Greene, D., & Shenoy, J.
Year & Journal	See Accepted_paper_code_to_ms.do

Data Structure

Field	Value
Data type	Panel firm-year (US firms, ~1990s-2010s)
Observations (N)	13,247 (after sample filter: census_region_south notna & randsamp1 notna)
Preferred model column	Table 4, Operating Profitability

Core Regression Model

Field	Value
Dependent variable	oibdp_atw (operating income before depreciation / assets, winsorized)
Focal IV	ad_law2 (anti-discrimination law indicator)
Controls	ln_at_adj, ppent_at, div_payer, state_inc_growth
Fixed effects	cm_id (firm) + fyear (year) + i(state_id, fyear) state-year trends
Clustered SEs	CRV1 clustered on cm_id (firm)

Simulation Configuration

Field	Value
Key variables (3)	ln_at_adj, ppent_at, state_inc_growth
MAR control	lev_lt_at
Mechanisms	MCAR, MAR, NMAR
Missingness proportions	1%, 5%, 10%, 20%, 30%, 40%, 50%
Methods	LD, Mean, Reg, Iter, RF, DL, MILGBM
Iterations per scenario	30
MAR/NMAR strength	1.5
M (multiple imputation)	5

Baseline Validation

Field	Value
Baseline coefficient	-0.0110 (ad_law2)

Field	Value
Baseline SE	0.0034
Baseline p-value	0.0013
Published value	ad_law2 negative significant (Table 4, Operating Profitability)
R-squared	N/A (absorbed by FE)

Run Log

Field	Value
Full run start	2026-04-10
Full run complete	2026-04-11
Total combinations	13,230 (3x7x3x7x30)
Errors	313 (A singular matrix: i(state_id,fyear) rank-deficient at high missingness; 2.4% rate, expected)
DL method	TensorFlow 2.21.0, MLP 32->Dropout(0.1)->16->1, ReLU, Adam lr=0.005, EarlyStopping
MILGBM method	LightGBM 4.6.0, MICE M=5, Rubin's Rules, 30 estimators, max_depth=4
Note	State-year linear trends via i(state_id, fyear) become rank-deficient at high missingness levels with LD